

STEP 6 > SMT ON PCB BOARDS

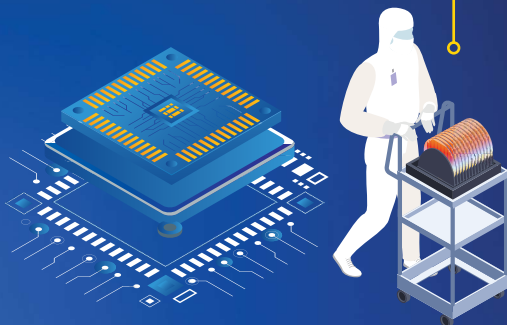
Bringing together chips, components, and modules to deliver finished devices to market

Global leaders: Flex, Jabil

**STEP 7 > FINAL PRODUCT ASSEMBLY**

Integrating components to manufacture end products such as smartphones and electronics.

Global leaders: Foxconn, Pegatron



design automation (EDA) companies.

Indian conglomerates are also recognising that they need chip design in their portfolios to build advanced systems that compete globally, he says.

Chip design is a long-haul effort, often requiring multiple iterations before success, with each adding to research and development (R&D) costs. Once a design is finalised, companies rely on overseas foundries such as TSMC and United Microelectronics Corporation for manufacturing, resulting in long wait times between tape-out and production.

Until recently, chips designed in India were mostly fabricated in Taiwan, and tested and packed elsewhere, before being shipped back to India or to end-customers. The absence of domestic fabs and packaging facilities meant higher costs, longer cycles and limited bargaining power. This was an unviable equation in a highly cost-competitive market, say Kanishka Chauhan, Senior Principal Analyst at Gartner.

Without a fab, it was not viable for Indian companies to change their business model from design as a service company to fabless, or one that designs semiconductors and outsources production. Recent fab announcements in India are beginning to alter that by bringing the ecosystem closer home.

“Companies are no longer competing for capacity from a position of weakness, nor absorbing the cost of chips moving across multiple geographies for fabrication and packaging,” says Chauhan.

MANAGING RISK

Semiconductor chip design is a business that rewards patience and deep capital. It demands sustained investment in EDA tools, IP libraries, compute infrastructure and large engineering teams. Multiple design iterations, failed tape-outs, and long development cycles are common, making chip design a high-risk exercise well before manufacturing begins.

“Designing a simple power chip can cost \$5–10 million for one product. But if you take a complex chip, such as a GPU (graphics processing unit) or a CPU (central processing unit), the investment can be between \$200 million and even half-a-billion dollars, when you factor in tape-outs, iterations, software and validation,” says Sridhar Chinamilli, Co-founder & CEO of Tessolve Services, owned by Hero Electronix.

In an industry that demands sustained capital and patience, most do not survive that journey; industry estimates say nine out of ten semiconductor start-ups fail before reaching scale.